LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	20
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Description purpose typical pneumatic system components
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Description purpose typical pneumatic system components
5. Specific Objectives	17.1.4.1 Trainee will be able to describe the construction and functioning of a typical pneumatic system and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of pneumatic system schematic diagrams; Instructor supervision during diagram analysis.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary source of compressed air in most aircraft?
(b) Name one component used to store compressed air.
(c) Why is pneumatic power preferred for certain aircraft systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and purpose of pneumatic systems • Air generation methods: APU and engine bleed • Air storage and regulation components • Distribution and control valves • Safety relief and monitoring devices	L & Ds	PPT, IB	15 min
(b)	• Detailed construction of air compressors • Functioning of air conditioning packs • Operation of pneumatic power units • Role of start valves and bleed valves • Maintenance requirements for pneumatic components	L & Ds	PPT, PDF	20 min
(c)	• Troubleshooting common pneumatic faults • Interconnection with other aircraft systems • Impact of system failures on aircraft performance • Standard operating procedures for pneumatic systems • Safety protocols during pneumatic system checks	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the function of a pneumatic control valve in a typical system.
(b) Describe the sequence of air flow from the engine to the cabin.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between engine bleed air and APU air for the next lesson on Aircraft controls systems.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Aircraft controls systems; mechanical, power assisted, fully powered & flies by wire type of controls surfaces.	Refer to the master syllabus reference list	LP 21

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min